PASHMEEN KAUR

B.Tech, Chemical Engineering · National Institute of Technology, Raipur · Graduating 2027

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EDUCATION

National Institute of Technology, Raipur — B.Tech, Chemical Engineering — 2027 | CGPA: 8.4 / 10

Surya Public School, Lucknow — CBSE Class XII — 2023 | 89.4%

St. Mary's Convent Inter College, Lucknow — ICSE Class X — 2021 | 96.4%

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML5, CSS3

Computer Vision: OpenCV, NumPy, YOLOv11, Roboflow, CVAT, image annotation & dataset labelling, camera calibration, planar homography, thresholding & segmentation, ROI / zone definition, morphological operations, measurement validation

AI & ML: RAG, LangChain, LangGraph, OpenAI API, Hugging Face, LLM fine-tuning, Prompt Engineering, scikit-learn, model training & evaluation (precision / recall, error analysis)

Backend & APIs: FastAPI, Flask, Node.js, Uvicorn, REST API design, HTTP methods, Postman

Frontend & Design: React.js, Next.js, React Native (Basics), Tailwind CSS, shadcn/ui, Framer Motion, Figma, UI/UX Design

Databases: MySQL, MongoDB

Cloud & DevOps: Docker, Git, GitHub Actions (CI/CD), Vercel, Render, Netlify

Product: Problem discovery with domain SMEs, requirement gathering from on-site field visits, MVP scoping, edge-case analysis, demo & stakeholder presentation, technical documentation

EXPERIENCE

Software Engineering Intern, Frontend Web Development — Cloord Solutions

Jun – Jul 2026

Next.js, React, TypeScript, Tailwind CSS, shadcn/ui, Framer Motion, Git, GitHub

- Built a responsive Next.js/React landing page in TypeScript with Tailwind CSS, shadcn/ui and Framer Motion, shipping reusable Navbar, Hero, Dashboard Preview, Features, CTA and Footer components.
- Implemented responsive layouts, animations, interactive states and dashboard visualisations aligned to the product design system.
- Integrated and maintained features within an existing production codebase, collaborating through Git feature branches, pull requests and code review.

PROJECTS

Hira Vision — Industrial Grate Wear Inspection (Hira Group, Raipur · Best Team Award)

GitHub

Python, OpenCV, NumPy, FastAPI, Uvicorn, Roboflow, YOLOv11, CVAT, JavaScript, HTML/CSS, Git

- Built an end-to-end computer-vision system measuring grate-slot wear in millimetres on a rotating drum screen from existing plant CCTV, replacing manual caliper inspection that required shutting the machine down and sending a worker inside.
- Scoped the problem through on-site plant visits and iterative review with Hira Group subject-matter experts, converting engineering drawings and maintenance tolerances (4 mm nominal / 5 mm watch / 6 mm reject) into measurable acceptance criteria.
- Calibrated pixels to millimetres via planar homography onto a 315×255 mm reference plane, then validated that calibration against an independent drawing dimension it never used — measured slot length landed within 3% of spec, making every reading self-checking.
- Benchmarked a YOLOv11 detector trained in Roboflow on 378 slot instances hand-annotated in CVAT; per-hole detection recovered only 20–32 of 39 slots under glare and debris, with recall varying unpredictably per plate.
- Replaced it with an open-area estimator immune to missed detections, cutting plate-to-plate spread from 1.24 mm to 0.86 mm and frame-to-frame repeatability to ±0.35 mm; chose the explainable pipeline over a black box so each maintenance decision stays auditable.
- Shipped a FastAPI + JavaScript web application: drag-and-drop upload, camera fingerprinting that auto-loads saved ROI zones per angle, interactive zone annotation for new camera positions, live annotated video with colour-coded verdicts, and a downloadable bundle of per-plate CSV, per-frame log, evidence images and report.
- Hardened for production edge cases — glare suppression via global thresholding, belt-speed tracking by 1-D profile correlation, and partial-plate frame trimming that reduced per-plate deviation from 2.40 mm to 0.82 mm.

Enterprise Multi-Agent RAG Support Agent

GitHub

Python, Flask, OpenAI API, ChromaDB, LangChain, LangGraph, REST APIs, React.js, Tailwind CSS

- Designed an end-to-end RAG architecture with LangChain and LangGraph, combining semantic embeddings and vector search over ChromaDB to answer context-aware queries grounded in enterprise document stores.
- Built a multi-agent orchestration layer with agentic tool calling and prompt engineering, routing multi-step support workflows across cross-platform endpoints.
- Developed modular Flask REST APIs as the agent backend, with structured request/response contracts and isolated failure handling per tool call.
- Engineered streaming responses, async API state and graceful degradation during high-latency LLM calls, keeping the UI responsive under slow inference.

Lead CRM Dashboard

Live Demo · **GitHub**

React.js, TypeScript, Tailwind CSS, Node.js, REST APIs, @dnd-kit, React Router DOM, Figma

- Built and deployed a responsive CRM web app in React.js and TypeScript, translating high-fidelity Figma designs into a reusable,

modular component library.

- Engineered Kanban drag-and-drop pipelines with @dnd-kit, using optimistic UI updates, client-side state validation and loading skeletons to keep interactions instant under network latency.
- Integrated asynchronous REST APIs for full CRUD operations, dynamic status filtering and real-time data sync, with component-level error boundaries and retry-safe failure handling.
- Enforced strict TypeScript typing and rule-driven state-transition guards to make invalid lead states unrepresentable, eliminating a class of data-integrity bugs.

ACHIEVEMENTS

- **Best Team Award — Future Ready Cross Business Garh**, in partnership with Hira Group, Raipur: progressed through build and demo review rounds to deliver a validated, industry-grade MVP running on live plant CCTV, developed with on-site plant visits and continuous subject-matter-expert feedback.
- **National Finalist** — top 6 of 10,000+ participants, Ground Business Reality 2026, BITS Pilani (Healthcare domain): designed a Closed-Loop Insulin Dosage Optimizer and delivery simulation.
- **First Runner-up** — UI/UX Challenge, Xccelerate by Money Matters with CSIDC 2026, IIIT Naya Raipur: built POCKETWISE, an investment portfolio recommendation screen with predictive modelling for savings and expense tracking.
- **Award of Excellence and Certificate of Merit**, CISCE — 96.4% in the ICSE examination, 2021.

COURSEWORK & CERTIFICATIONS

- The I-Scale: Complete AI Engineer Full Course — AI/ML in 16 Hours · Google: Intro to Generative AI (certification)
- **AI Agents:** Coding agents (Cursor, Hermes, Open Code, Antigravity, Claude Code); Automation (Composio, Claude, Gemini, Zapier, n8n, Lovable); Agents & Workflow management (Google AI Studio, Notebook LLM, Deep Seek, Nano Banana); LLMs (ChatGPT, Gemini, Perplexity, Claude)
- **Relevant Coursework:** Generative AI, AI/ML Fundamentals, AI Agents for Automation, Large-Language Models (LLMs), Statistics for Data Science

POSITIONS OF RESPONSIBILITY

Frontend Technical Executive — Innovation Cell, NIT Raipur Jun 2024 – Aug 2025

- Maintained and shipped updates to the official Innovation Cell website, owning UI consistency and release cadence.

PR & Sponsorship Coordinator — Literati, NIT Raipur 2023 – Present

- Coordinated and anchored flagship literary events end to end; authored articles for Shilpi, the annual magazine (2023–2025).